PROJECT SUBMISSION

ACADEMIC INNOVATION

1. TwinSense AI

Predictive Healthcare Digital Twin Prototype for Continuous Glycemic Risk Estimation

In-Silico Simulation

Core Digital Twin Concept

- In-Silico Patient Avatar: Unifies static EHR baselines with high-frequency dynamic CGM telemetry.
- 45-Minute Horizon: Forward-looking risk classification (In-Range vs. Hyperglycemia).
- Counterfactual Sandbox: Non-destructive 'What-If' lifestyle and activity simulation.
- Explainable AI: Automated physiological risk factor attribution.
- Local-First Privacy: 100% offline-capable, zero external cloud dependencies.

100% Synthetic Data

Mandatory Ethical & Clinical Disclaimer

- FOR RESEARCH & DEMONSTRATION PURPOSES ONLY.
- **100% Synthetic Data: All profiles and CGM time-series are algorithmically generated.**
- **Not Clinically Validated: Not evaluated in clinical trials; NOT approved by FDA/CE/CDSCO.**
- Zero Medical Claims: Must NEVER be used for diagnostic or insulin dosage decisions.
- Unsupported Hypoglycemia: Model does not detect or predict hypoglycemia (Class 1).

CLINICAL CONTEXT

THE CHALLENGE

2. Problem Statement & Clinical Motivation

The challenge of glycemic volatility and the limitations of current CGM alert systems

Clinical Urgency

Glycemic Volatility in Diabetes

- Dynamic Instability: Individuals with Type 1 & Type 2 diabetes suffer rapid glycemic excursions.
- Acute Risks: Hyperglycemic crises (ketoacidosis, hyperosmolar state) vs. acute hypoglycemic shock.
- Multi-Factorial Drivers: Volatility is driven by delayed meal absorption, physical exertion, and circadian shifts.
- Action Delay: Corrective lifestyle or dietary actions are often taken too late after symptoms arise.

Point-in-Time Lag

Standard CGM System Limitations

- Point-in-Time Readings: Existing CGMs report current sensor values and basic trend arrows, not 45m forward risk.
- No Behavioral Sandbox: Patients cannot safely test 'what if I walk 1,000 steps?' in advance.
- Black-Box Alerts: Standard threshold alarms lack explainable contributing physiological factors.
- Need for In-Silico Testing: A software digital twin provides an exploratory space without biological risk.

SOLUTION OVERVIEW

DIGITAL TWIN

3. Proposed Solution: TwinSense AI

Coupling physiological EHR baselines with wearable telemetry for predictive risk estimation

TwinEngine Core

The Digital Twin Architecture

- In-Memory Representation: Combines patient EHR profiles (age, BMI, HbA1c) with real-time CGM telemetry.
- 45-Minute Forward Inference: Classifies future glycemic excursion risk into defined clinical categories.
- Contributing Factors Panel: Breaks down risk into 15m rate of change, rolling means, and meal events.
- Strict Temporal Boundaries: Validates chronological ordering and enforces a minimum 30-minute sensor history.

Four Pillars

Core Capabilities Implemented

1. 45-Min Forward Risk Classification: Balanced Random Forest trained with a 45m temporal embargo.
2. What-If Counterfactual Sandbox: Ephemeral in-memory clone evaluating hypothetical activity shifts.
3. Independent Rule-Based Low Glucose Alert: Immediate trigger when measured reading is strictly < 70 mg/dL.
4. Downloadable Audit Reports: Clinical summary PDF and tabular CSV generated directly via ReportLab.

APPLICATION DOMAIN

DECISION SUPPORT

4. Healthcare Use Case & Target Setting

Decision support, behavioral exploration, and clinical research prototyping

T1D & T2D

Target Patient Population

- Type 1 & Type 2 Diabetes: Focus on individuals utilizing contin uous glucose monitoring devices.
- Clinical Diversity: Evaluated across diverse synthetic profiles (ages 28–64, BMI 22.4–34.2, HbA1c 6.7%–10.0%).
- High-Frequency Telemetry: CGM sampled at 5-minute continuous in tervals alongside heart rate and step counts.
- Safe Research Testbed: Provides an academic benchmark for physi ological modeling and digital twins.

Proactive Care

Clinical & Behavioral Value Drivers

- Proactive Lead Time: 45-minute forward window allows early hydr ation, resting, or light activity.
- Safe Lifestyle Experimentation: Users evaluate hypothetical ste p increments without physical danger.
- Explainable Consultations: PDF clinical reports summarize risk probability distributions for physician reviews.
- Pure Non-Clinical Guardrails: Never provides insulin dosing; ma ndates physician consultation.

SYSTEM ENGINEERING

TWO-PHASE DESIGN

5. End-to-End System Architecture

Decoupled batch training pipeline and low-latency local runtime microservice

Data & ML

Phase 1: Offline Batch Pipeline

- Synthetic Generator: src/data_generator.py creates raw patient profiles and 20,160 CGM readings.
- Feature Pipeline: src/feature_engineering.py computes past-only rolling metrics and drops lookahead leaks.
- Model Training: src/train.py partitions data with a 45m embargo buffer and trains Random Forest.
- Artifact Serialization: Persists models/random_forest_risk_model.joblib and feature_columns.json.

Local ASGI Server

Phase 2: Online Runtime Sandbox

- Client Dashboard: Vanilla HTML5, CSS3, ES6 JS with pure native SVG telemetry charting (zero CDNs).
- FastAPI Backend: Typed asynchronous REST API (port 8000) with Pydantic v2 schemas.
- Local Relational DB: SQLite3 (data/twinsense.db) storing patients, telemetry, audit logs, and sessions.
- Security: RFC 9106 Argon2id password hashing, HttpOnly session cookies, and CSRF protection.

DATA SCIENCE

LEAK-FREE PIPELINE

6. Synthetic Dataset & Feature Engineering

Deterministic simulation, past-only rolling features, and the 45-minute embargo protocol

100% Synthetic

Deterministic Synthetic Generator

- Multi-Patient Cohort: 10 clinically diverse synthetic patients (SYNTH_PAT_001 to 010).
- Longitudinal Scope: 7 continuous simulated days at 5-minute sampling intervals (20,160 total rows).
- Physiological Bound Validation: Strictly checked [40.0, 400.0] mg/dL, strictly monotonic timestamps.
- Zero Real-Patient Data: Generated via mathematical circadian and postprandial clearance heuristics.

Embargo Buffer

11-Dimensional Feature Pipeline

- Rolling Glucose Features: 15m and 30m rolling means, 15m rate of change (mg/dL per 15 min).
- Dynamic Activity Telemetry: 30-minute accumulated step count, 30-minute mean heart rate.
- Static EHR Phenotype: Age, BMI, baseline HbA1c, diabetes type (T1D/T2D), and diagnosis duration.
- 45-Min Embargo Protocol: Drops 9 consecutive samples at split boundaries to eliminate temporal leakage.
- Processed Dataset: 20,020 leak-free feature-target pairs in data/processed/training_dataset.csv.

MACHINE LEARNING

45-MIN HORIZON

7. Random Forest Model & 45-Minute Prediction

Balanced multi-class ensemble modeling evaluated against an unlearned majority baseline

Random Forest

Model Architecture & Training

- Algorithm: Balanced RandomForestClassifier (n_estimators=100, max_depth=8, class_weight='balanced').
- Chronological Partitioning: First 80% of patient timelines for training, final 20% for untouched testing.
- Benchmark Comparison: Benchmarked against an unlearned DummyClassifier(strategy='most_frequent').
- Feature Schema Enforcement: Strict 11-feature ordered contract guaranteed via feature_columns.json.

3 Target Classes

Clinical Risk Class Definitions

- **Class 1: Hypoglycemia (< 70 mg/dL) — CRITICAL LIMITATION: 0 samples in synthetic data.**
- Class 2: In-Range / Normal Glycemia (70 – 180 mg/dL) — 1,756 test samples.
- Class 3: Hyperglycemia (> 180 mg/dL) — 2,164 test samples.
- Prediction Output: Calibrated categorical probabilities, forward target timestamp (+45m), and factors.

BENCHMARK METRICS

HONEST DISCLOSURE

8. Actual Evaluation Results & Limitations

Documented benchmark performance and transparent disclosure of the hypoglycemia limitation

89.29% Accuracy

Held-Out Test Set Performance (N=3,920)

- **Overall Accuracy: 89.29% (vs. Majority Baseline: 55.20%, +34.08 % improvement).**
- **Macro F1-Score (Active Classes): 0.8917 (vs. Majority Baseline: 0.3557, +0.5360 improvement).**
- Class 2 (In-Range): Precision 88.00%, Recall 88.10%, F1-Score 0 .8805 (1,547 correct / 1,756).
- Class 3 (Hyperglycemia): Precision 90.33%, Recall 90.25%, F1-Sc ore 0.9029 (1,953 correct / 2,164).
- Validated on untouched chronological holdout data with a 45-min ute embargo buffer.

Zero Support

The Hypoglycemia (Class 1) Limitation

- Zero Class 1 Samples: Because synthetic data had elevated basel ines, 0 hypoglycemia events were simulated.
- Unsupported Model Scope: The model CANNOT detect, predict, or r ule out hypoglycemia.
- Honest UI Policy: Class 1 probability is explicitly displayed a s 'Not supported' (NEVER '0%').
- Warning Banner: 'Hypoglycemia prediction is not supported by th e current model' in all reports & panels.
- Pure Synthetic Data: Metrics reflect synthetic simulation and D O NOT imply clinical efficacy.

FRONTEND & UX

LIVE DASHBOARD

9. Dashboard Interface & Implemented Features

Native web client, real-time SVG charting, and full audit trail capabilities

Zero Dependencies

Interactive Client Capabilities

- Patient Phenotype Selector: Switch between 10 synthetic patient s; profiles update dynamically.
- Native SVG CGM Chart: Mathematical coordinate renderer with col or-coded target zones (zero chart.js).
- 45-Min Forward Forecast Panel: Visual risk badges, probability distributions, and clinical factors.
- What-If Counterfactual Controls: Interactive sliders for Delta Glucose and Delta Activity.
- Low-Glucose Safety Alert: Prominent banner fires immediately if measured glucose is < 70 mg/dL.

Enterprise Ready

Backend Services & Audit Trail

- Argon2id Security: Session-based login/registration with secure HttpOnly cookies and CSRF tokens.
- Downloadable PDF Report: Professional clinical summary PDF gene rated via ReportLab.
- Downloadable CSV Report: Full tabular audit export containing t imestamps, features, and probabilities.
- Historical Predictions API: SQLite audit trail tracking every p rediction run for traceability.
- Live Web App: Accessible locally at http://127.0.0.1:8000/ with interactive Swagger docs at /docs.

SAFETY & SANDBOX

GUARDRAILS

10. Safety Alerts & What-If Simulation

Rule-based emergency protection and non-destructive counterfactual exploration

Strictly Rule-Based

Independent Low-Glucose Safety Alert

- Trigger: Fires strictly when measured observed glucose reading is < 70 mg/dL.
- Independent of ML: Activates immediately upon patient selection without running a model forecast.
- Measured Readings Only: Hypothetical what-if slider values NEVER trigger this alert.
- **Strict Safety Wording: Advises immediate medical help if symptoms occur; ZERO dosing terms.**
- Reset on Patient Switch: Automatically updates or clears when switching patient records.

In-Memory Copy

What-If Counterfactual Sandbox

- Ephemeral Telemetry Clones: Modifies an in-memory clone without mutating database or CSVs.
- Evaluates Dual Perturbations: Delta Glucose (-50 to +50 mg/dL) and Delta Steps (0 to +1,500).
- Side-by-Side Comparison: Visualizes observed baseline vs. simulated hypothetical forward risk.
- Clear Simulation Labeling: Hypothetical values carry a prominent '(Simulated)' badge.
- Zero Reassurance: Maintains all disclaimers and displays Class 1 as 'Not supported'.

ENGINEERING RIGOR

123 TESTS PASSING

11. Technology Stack & Verification Suite

Production-grade Python ecosystem validated by 123 automated test cases

Modern Python

Core Technical Stack

- Backend & API: FastAPI >=0.110.0, Uvicorn >=0.28.0, Pydantic v2 >=2.6.0.
- Machine Learning: Scikit-Learn >=1.4.0, Pandas >=2.2.0, NumPy >=1.26.0, Joblib >=1.3.0.
- Security & DB: Argon2-cffi >=25.1.0, Python Built-in SQLite3, CSRF Protection.
- Reporting: ReportLab >=4.0.0 (Clinical PDF Synthesis), pypdf >=5.0.0.
- Frontend: Semantic HTML5, CSS3 Variables, ES6 JavaScript, Pure Native SVG.

100% Pass Rate

Verification & Quality Assurance

- Pytest Suite: 123 automated test cases covering API, TwinEngine, Auth, Safety, and Reports.
- Immutability Checks: Tests confirm what-if simulations leave underlying state unaltered.
- Embargo Verification: Asserts zero temporal target leakage across chronological splits.
- Safety Unit Tests: Asserts boundary alerts (69.9 vs 70.0 mg/dL) and absence of dosing terms.
- Cryptographic Integrity: SHA-256 hashes of datasets and models validated bit-for-bit unchanged.

SUMMARY & FUTURE

CONCLUSION

12. Conclusion & Future Research Roadmap

Key contributions of the TwinSense AI prototype and next engineering frontiers

Validated Prototype

Key Project Achievements

- Proof-of-Concept Digital Twin: Proved feasibility of coupling E HR baselines with CGM telemetry.
- Leak-Free Machine Learning: Demonstrated rigorous 45-minute emb argo buffer implementation.
- Exploratory Sandbox: Delivered interactive, non-destructive cou nterfactual what-if simulation.
- Safety by Design: Built decoupled rule-based low-glucose alerts and transparent limitation notices.
- Enterprise Local Architecture: Implemented full Argon2id authen tication and ReportLab PDF reporting.

Next Steps

Future Research Horizons

1. Hypoglycemia Simulation: Model missed meals, insulin stacking, and strenuous exercise.
2. Open Simulators: Integrate with established UVA/Padova simulat ors (e.g., simglucose).
3. Real Clinical Benchmarking: Validate against de-identified cli nical datasets (OhioT1DM / Tidepool).
4. Deep Recurrent Architectures: Experiment with Temporal Fusion Transformers (TFT) and LSTMs.
5. Pharmacokinetic Modeling: Incorporate active insulin-on-board (IOB) clearance equations.